AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 2 · STUDENT LESSON

# Features, Labels, Targets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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## Before you start

AI-2 Week 9: a feature is an input; the target is what is predicted.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Separate prediction inputs from observed targets. Try the recall before opening notes.

### Your learning target

I can separate **prediction input**s from observed **target**s.

I can check whether each **feature** exists at **prediction time**.

I can repair **target leakage** and state a **feature contract**.

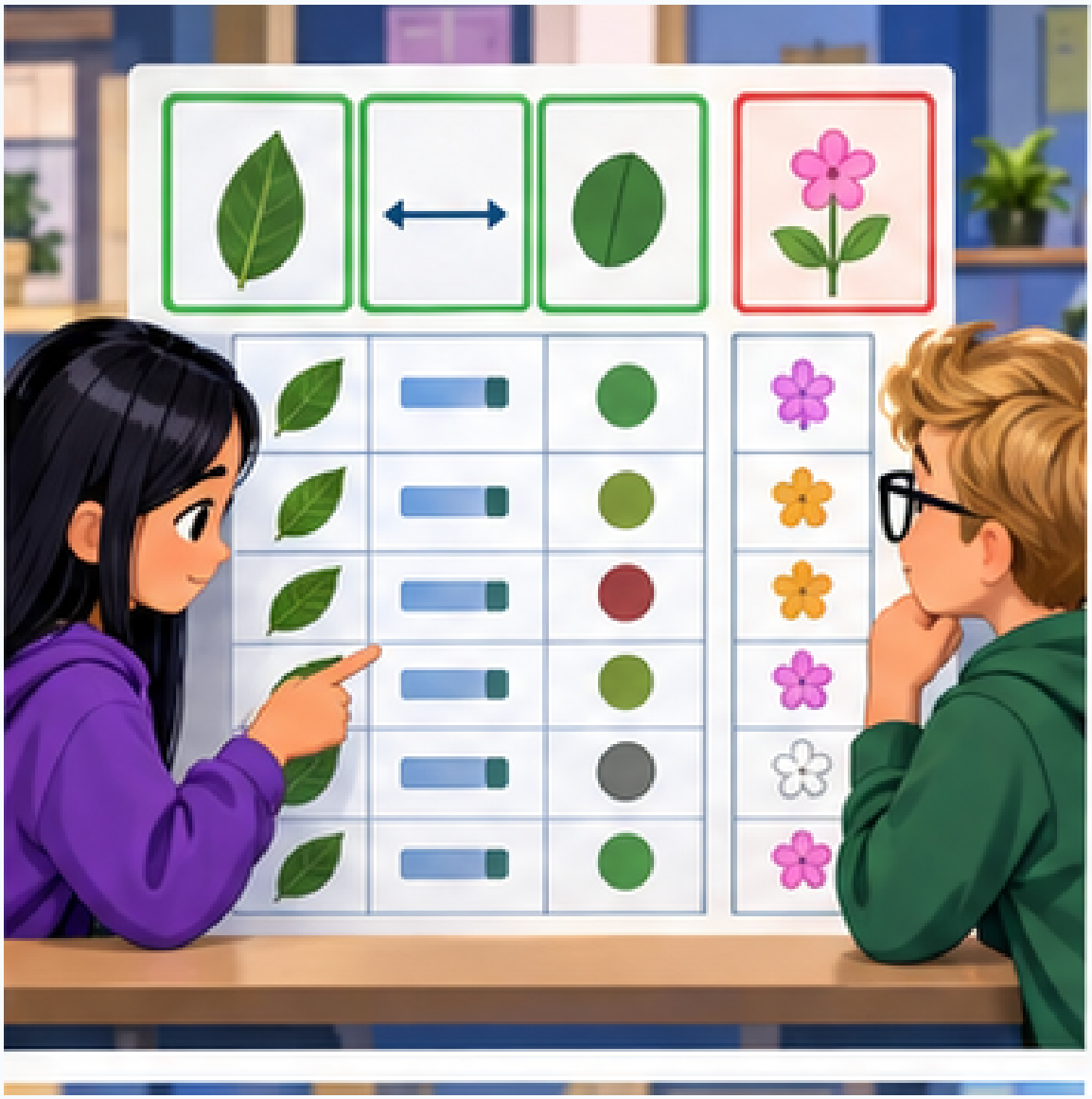
### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

**Feature**s are the information available at **prediction time**; the **target** is the value the model should estimate; **label**s are observed **target** values in examples.



### Your target

LEARNING **TARGET** A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

Builder name and experiment date:

### Readiness for the connected explanation

> Why must held out **label**s not be used to choose the threshold in Week 1?

### This week's end goal

By the end, I can explain **feature**s, **label**s, **target**s using a traced example and a stated limitation.

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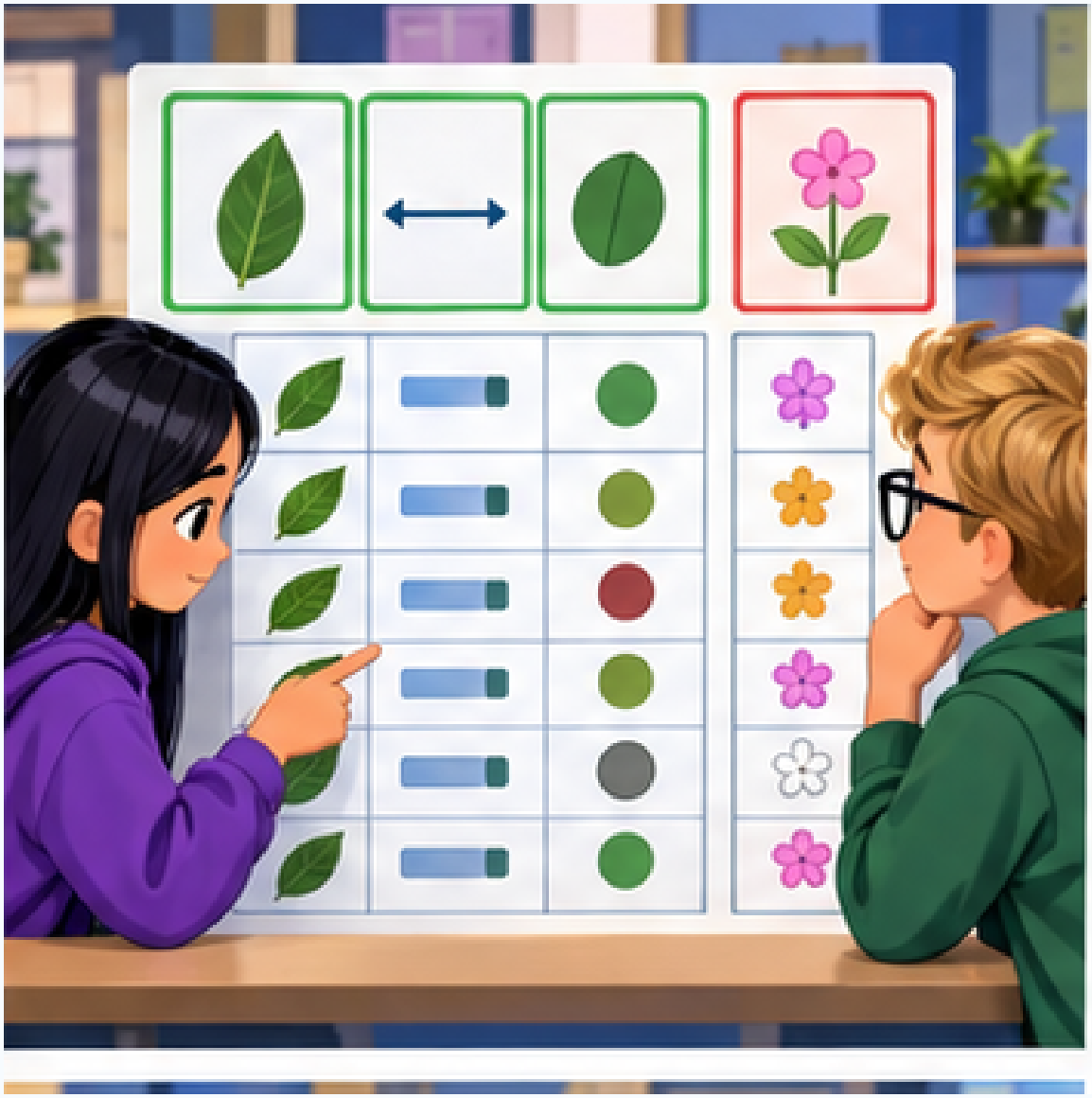
## Notice the evidence

> Investigate the supplied Features, Labels, Targets case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

### Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



### Start with this puzzle

MYSTERY The team must use **feature**s, **label**s, **target**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

### A closer explanation

Week 1 separated **training** examples from testing examples. This week asks a different question: could each **input** actually be known when a **prediction** is needed? A historical table may contain facts added after the event.

A model trained with those facts can appear successful even though the **input**s will be missing when it is used.

Use fictional library loans. At the instant a book is checked out, predict whether it will be returned more than 14 days later.

For completed historical loans, **label** 1 means days_to_return is greater than 14; **label** 0 means it is at most 14. Exactly 14 days is on time. Assume no changed due dates in this exercise.

All records and values are invented; do not collect classmates' borrowing histories.

| Field | When known | Role for this task |
|---|---|---|
| pages | At checkout | Candidate feature |
| queue_at_checkout | At checkout | Candidate feature |
| days_to_return | After return | Creates the label only |
| returned_late | After return | Label to predict |
| loan_id | At checkout | Tracking metadata |

queue_at_checkout is a saved count of waiting requests at that instant. It is not the queue when we later export the table. Candidate means available to consider, not proven useful or appropriate.

Exclude the arbitrary loan identifier from this exercise; keep it separately to track records.

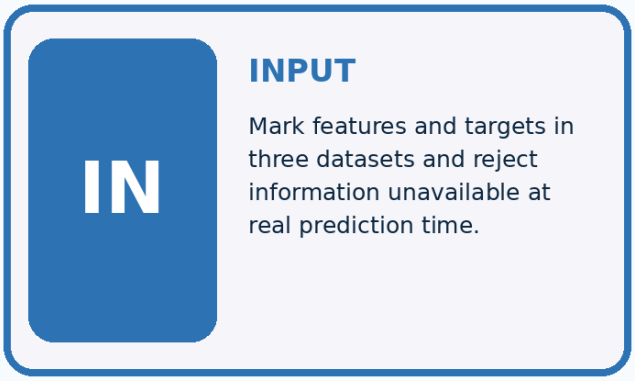
## See how it works

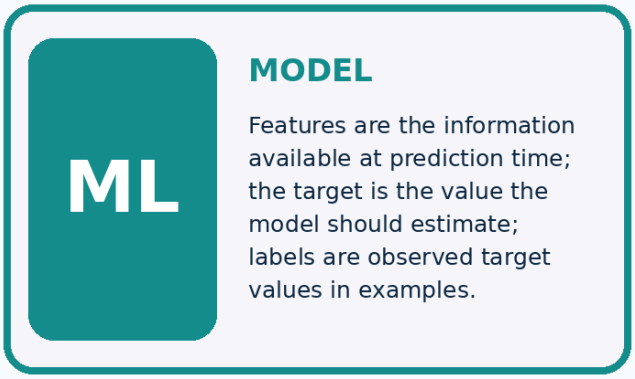
> Follow the mechanism for Features, Labels, Targets. Connect each labelled input, deciding step and result.

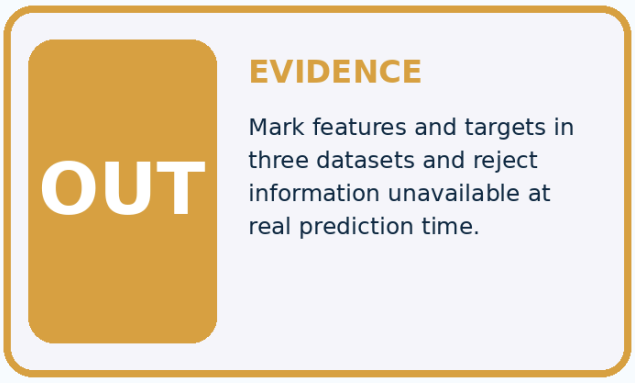
ML MECHANISM

**Input → Model → Evidence**

Trace the information, fitted behavior, and evaluation proof separately.






### Question to explain

QUESTION Mark **feature**s and **target**s in three datasets and reject information unavailable at real **prediction time**.

MODEL IDEA **Feature**s are the information available at **prediction time**; the **target** is the value the model should estimate; **label**s are observed **target** values in examples.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

### How the mechanism works

UNDER THE HOOD Plant row: leaf_width=6 cm, color_score=0.72, health=spotted. **Feature**s are leaf_width and color_score because they exist before **prediction**.

Draw a fresh data → model → **prediction** → evaluation trace:

### A closer explanation

```
def features(row):
    return [row["pages"], row["queue_at_checkout"]]

past = {"loan_id": "A", "pages": 120,
        "queue_at_checkout": 2, "days_to_return": 18,
        "returned_late": 1}
print(features(past))
```

```
print(past["returned_late"])
new = {"loan_id": "B", "pages": 80,
       "queue_at_checkout": 0}
print(features(new))
```

The **feature** order is always pages first, queue count second. The first **output** is [120, 2]; the separate **label** is 1 because this historical loan took 18 days.

The new loan produces [80, 0] even though its return outcome does not yet exist. The same **feature** function serves past and new records.

It deliberately selects allowed fields rather than copying every column except returned_late.

### Follow information through time

| MOMENT | AVAILABLE INFORMATION | ALLOWED USE |
| --- | --- | --- |
| Checkout | Pages and saved queue | Prediction inputs |
| Later return | Elapsed days | Create observed label |
| Historical training | Inputs plus separate label | Fit a model later |

The outcome may teach the model during training, but it cannot be an input to an earlier checkout prediction.

This code prepares **input**s; it does not fit a model or predict lateness. It assumes the two **feature** fields contain valid nonnegative integer values and pages is positive. A missing required field raises KeyError.

A real pipeline needs explicit validation and a missing-data policy.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

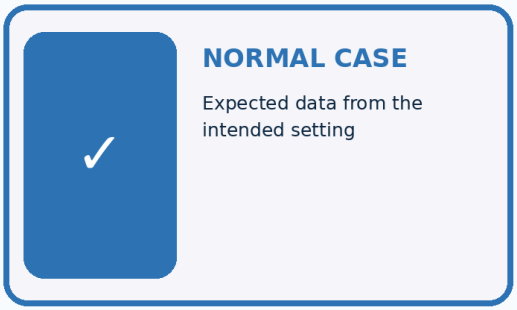
# Compare the cases

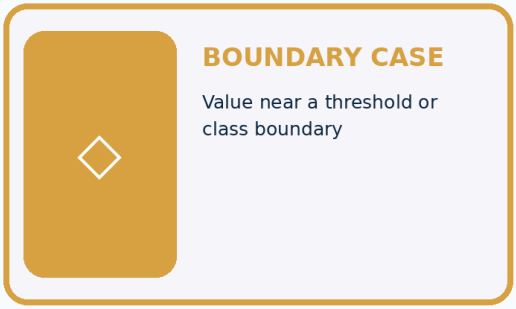
> Compare the supplied cases for Features, Labels, Targets. Identify the detail that changes the conclusion.

EVALUATION DESIGN

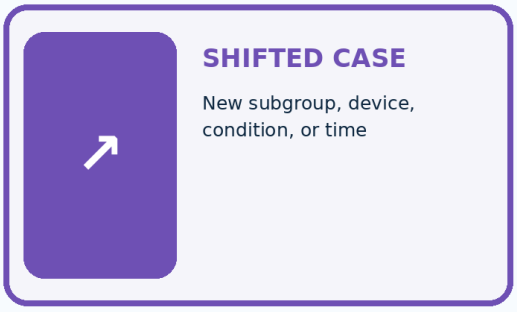
### Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.






NORMAL Expected data and expected behavior.

### Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous **label**, or uncertain **prediction**.

MISSING Necessary field, **label**, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

### Change one thing in the pictured case

#### Change one thing

Change: A table adds tomorrow_soil_moisture as an **input**.

#### Predicted result

**Prediction**: Reject it for this **prediction**-time contract.

#### Why the result follows

Reason: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

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## Words that unlock the idea

> Use the vocabulary of Features, Labels, Targets to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

### Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

**FEATURE**

input variable used for prediction

2

**TARGET**

quantity or category to estimate

3

**LABEL**

observed target value in a dataset

4

**ROW**

one example with fields

| Term | Precise meaning | My ML example |
|---|---|---|
| feature | input variable used for prediction | |
| target | quantity or category to estimate | |
| label | observed target value in a dataset | |
| row | one example with fields | |

MISCONCEPTION TO REPAIR Any column correlated with the answer is automatically a good **feature**.

Repair the claim using a definition, calculation, or test record:

### Words used in the closer explanation

| Word | Meaning |
|---|---|
| Feature contract | A specification of input meaning units order and timing |
| Target | The outcome a prediction aims to estimate |
| Prediction time | The moment when the prediction must be made |
| Target leakage | Predictor information that reveals the outcome but is unavailable at prediction time |

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STUDENT LESSON · 6 OF 18

## Follow a worked example

> Follow the worked trace for Features, Labels, Targets, predict a step, then compare with its explanation.

WORKED EVIDENCE

## Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

**WEEK 2 • WORKED EVIDENCE**

1. Plant row: leaf_width=6 cm, color_score=0.72, health=spotted.
2. Features are leaf_width and color_score because they exist before prediction.
3. Target is health; the stored value spotted is this row's label.
4. A treatment-result field recorded later is not a valid prediction-time feature.

### Step 1

STEP 1 Plant row: leaf_width=6 cm, color_score=0.72, health=spotted.

### Step 2

STEP 2 **Feature**s are leaf_width and color_score because they exist before **prediction**.

### Step 3

STEP 3 **Target** is health; the stored value spotted is this row's **label**.

### Step 4

STEP 4 A treatment-result field recorded later is not a valid **prediction**-time **feature**.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

**A closer explanation**

### Worked explanation · Features, Labels, Targets

For every proposed **input**, ask what it means, when it was recorded, and whether the same definition can be supplied at checkout. A field in the historical export is not automatically available at **prediction time**.

An **input** calculated from later information is also unavailable, even if its name does not mention the outcome.

**A completed loan has pages 200, queue_at_checkout 3 and days_to_return 14. Write the feature list and label. Someone adds overdue_fee calculated after return. Can it become a feature for the checkout prediction? Explain.**

**Learn from the worked case**

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Features, Labels, Targets

Choose **feature**s that exist when a plant **prediction** must be made.

### The facts we will use

> **Supplied facts**
>
> Predict next-week wilting at today's photo capture. Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/2

# Features, Labels, Targets

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Mark prediction time

The decision occurs today, before next week's outcome.

### 2. Select inputs

Today's light and soil readings are available features; plant_id identifies the plant.

### 3. Keep target separate

next_week_wilt supplies the training label, not an input available today.

**CHANGE ONE THING**

A table adds tomorrow_soil_moisture as an input.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Mark prediction time | The decision occurs today, before next week's outcome. |
| 2. Select inputs | Today's light and soil readings are available features; plant_id identifies the plant. |
| 3. Keep target separate | next_week_wilt supplies the training label, not an input available today. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

#### Keep the stated limit

Eligible fields are not automatically useful or causal. Measure their predictive value with a

valid split.

### Keep the stated limit

Read the **label**led picture: Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Keep the stated limit

Change one thing: A table adds tomorrow_soil_moisture as an **input**. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Features, Labels, Targets claim and explain a repair using the supplied evidence.

### Find the first error

ERROR DETECTIVE

**Compare → Diagnose → Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

**COMPARE • DIAGNOSE • REPAIR**

Tempting claim: Any column correlated with the answer is automatically a good feature. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

| Evidence record | Expected | Actual | Likely cause / next test |
|---|---|---|---|
| normal case | passes | | |
| boundary case | defined behavior | | |
| missing case | safe fallback | | |
| shifted case | measured limit | | |

CLAIM TO CHALLENGE Any column correlated with the answer is automatically a good **feature**.

Diagnose one cause and name the cleanest next experiment:

### A closer explanation

## Check the claim against the facts

A **feature contract** records field names, meaning, units, order and availability time. Here the **input**s are pages in pages, then queue_at_checkout in requests, both captured at checkout.

The **target** definition is returned more than 14 days later. Historical outcomes may be available for **training**, while new outcomes are unknown.

Never fill an unknown **label** with 0 merely because the book has not yet been returned.

### Check the claim against the facts

Availability alone does not establish predictive value, fairness, a causal effect or a reason to use a model. This lesson audits **input**s; later lessons evaluate models.

If an **input** cannot be collected at checkout, revise the design or specify a justified missing-**input** policy before use.

**Write a two-sentence input contract for this checkout task. Include the feature order and timing, then explain why an unknown future outcome must not be recorded as an on-time return.**

STUDENT LESSON · 8 OF 18

## Practise together

Practise Features, Labels, Targets with support, then explain the deciding step in your own words.

ML STUDIO

### Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.




### Experiment record

#### Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + **feature**s + model + settings + seed

METRICS + positive class + units + denominators

**PREDICTION**S + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

### A closer explanation

A learner removes returned_late but keeps days_to_return in the **input**s. That still reveals the answer: compare days_to_return with 14. Remove both the **target** and the later return measurement from **prediction input**s.

Also exclude a fee calculated after return or a renewal made after checkout. These are examples of **target leakage**, where unavailable outcome information enters the predictors.

Moving rows into a test set does not repair this problem if each test row still contains the later return measurement. Rebuild the **input**s to match checkout time and repeat development and evaluation honestly.

If earlier test results influenced changes, obtain a new final evaluation set as well.

### Practise with less help: Trace the feature contract

Use workbook W2 for the connected example "**Feature**s **label**s and **prediction time**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for **Feature**s **label**s and **prediction time**. Proposed explanation: "Every stored column is a **feature**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

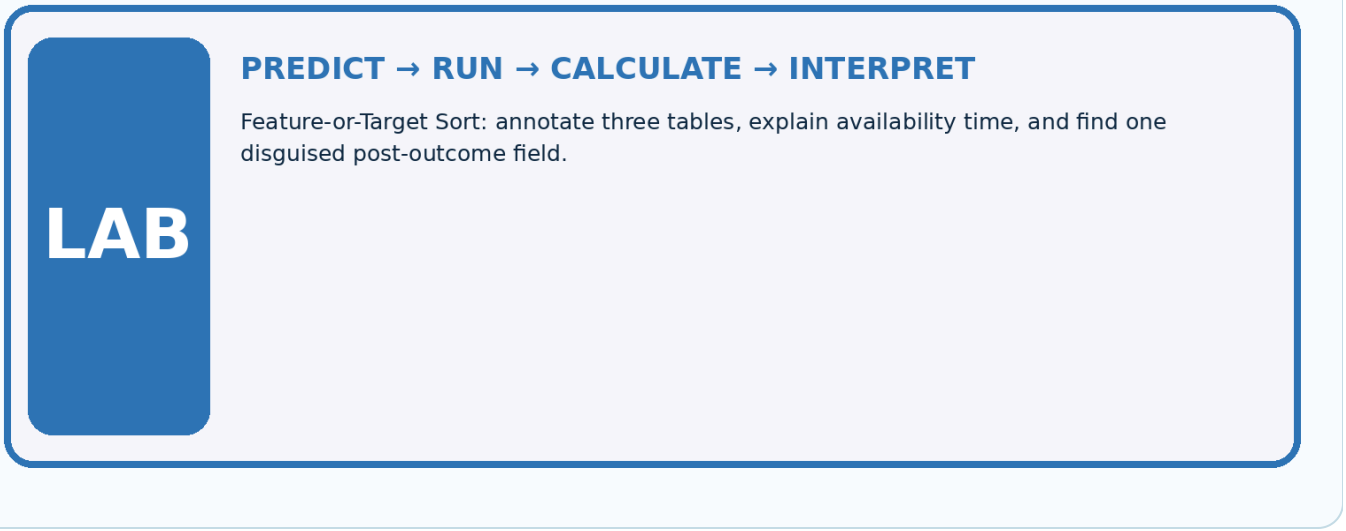
# Run the investigation

> Investigate Features, Labels, Targets. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

**Predict → Run → Calculate → Interpret**

Keep the test conditions fixed, record every result, and change only one planned factor.




1. Write the hypothesis and expected **evidence** before running.

2. **Feature**-or-**Target** Sort: annotate three tables, explain availability time, and find one

disguised post-outcome field.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

**Case:** Normal
**Prediction:** _____
**Actual:** _____
**Calculation / evidence:** _____
**Meaning / next test:** _____

**Case:** Boundary
**Prediction:** _____
**Actual:** _____
**Calculation / evidence:** _____
**Meaning / next test:** _____

**Case:** Missing
**Prediction:** _____
**Actual:** _____
**Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

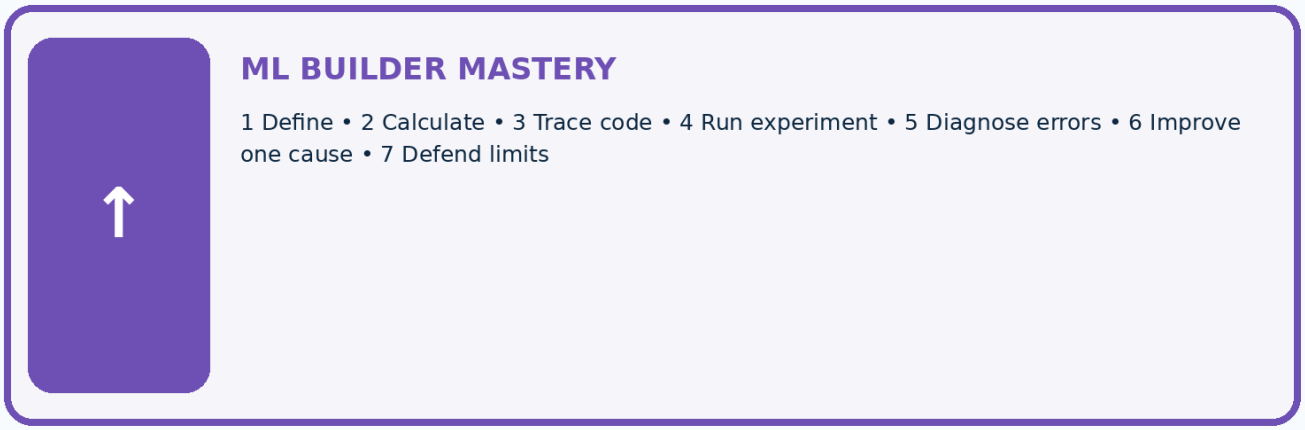
## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Features, Labels, Targets. Explain why each step is justified.

MASTERY

### From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.




DEFINE Use all four terms precisely: **feature**, **target**, **label**, row.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Any column correlated with the answer is automatically a good **feature**.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

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STUDENT LESSON · 11 OF 18

## Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

### Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.




THIS WEEK'S CHECKPOINT Define the Plant Classifier **target** and a first **feature** list; record units, allowed values, collection method, and availability time.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

### Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

**Label** this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

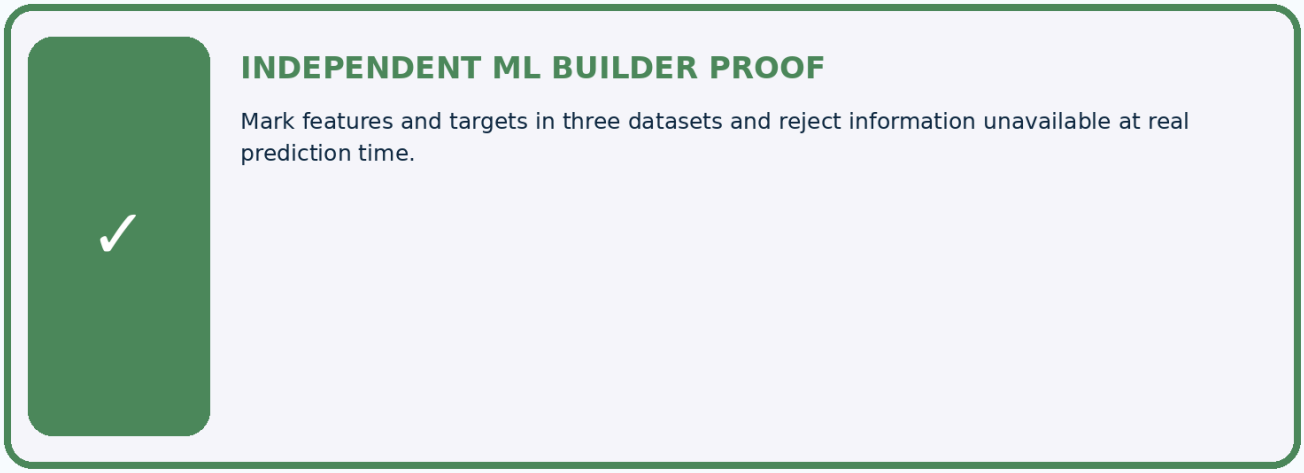
## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

**Explain • Calculate • Transfer • Defend**

Complete this with a fresh case after guided practice and compare it with the mastery criteria.




### What success looks like

REQUIRED PROOF Mark **feature**s and **target**s in three datasets and reject information unavailable at real **prediction time**.

- ☐ Define the task, **feature**/**target** roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.

> ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

### Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Classification or Regression?.

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STUDENT LESSON · 13 OF 18

## Study a complete case

> Revisit the complete Features, Labels, Targets case for supported practice. Repeated examples do not become fresh assessments.

### Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

Rows contain leaf length, edge type and healthy/spotted **label**.

### The question

Identify **feature**s, **target** and **prediction**-time **input**s.

### Worked explanation

Length and edge type are candidate **feature**s; the **label** is the **target**. Do not supply the true **target** as an **input** to predicting it. Evaluate **feature** usefulness.

### Explain it in your own words

> Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

### Trace the supplied case visually

### Separate features from the target

| Evidence or stage | Value or result |
|---|---|
| Length | Candidate input feature |
| Edge type | Candidate input feature |
| Healthy or spotted | Target label |
| Prediction time | True target is unavailable |

Rows contain leaf length, edge type and healthy/spotted **label**.

Identify **feature**s, **target** and **prediction**-time **input**s.

Length and edge type are candidate **feature**s; the **label** is the **target**. Do not supply the true **target** as an **input** to predicting it. Evaluate **feature** usefulness.

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Audit a new task independently

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two **feature**s and the **target**.

My answer, with a trace, calculation or supporting **evidence**

**Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

**Write or draw your reasoning and evidence.**

### Connect to your project

Define the Plant Classifier **target** and a first **feature** list; record units, allowed values, collection method, and availability time.

The **evidence** I will keep

**Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

### Transfer practice from the original programme

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

> **Explain the mechanism, evidence and one limit.**

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

> Use this map to navigate Features, Labels, Targets, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Choose **feature**s that exist when a plant **prediction** must be made.

| By the end, I can | How I will show it |
| --- | --- |
| Separate prediction inputs from observed targets | Explain the deciding step, show a trace or test, and state one limit. |
| Check whether each feature exists at prediction time | Explain the deciding step, show a trace or test, and state one limit. |
| Repair target leakage and state a feature contract | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Features, Labels, Targets without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Separate **prediction input**s from observed **target**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Choose **feature**s that exist when a plant **prediction** must be made.

### Practice 1: explain the deciding evidence

Which **label**led step matches this detail? next_week_wilt supplies the **training label**, not an **input** available today.

1. Select **input**s

2. Mark **prediction time**

3. Keep **target** separate

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep **target** separate next_week_wilt supplies the **training label**, not an **input** available today.

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: A table adds tomorrow_soil_moisture as an **input**. Which conclusion is supported by the supplied facts?

1. Reject it for this **prediction**-time contract.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject it for this **prediction**-time contract. Tomorrow's measurement is unavailable when today's **prediction** must be issued.

### Repair a missing building block

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

### Keep your completed work

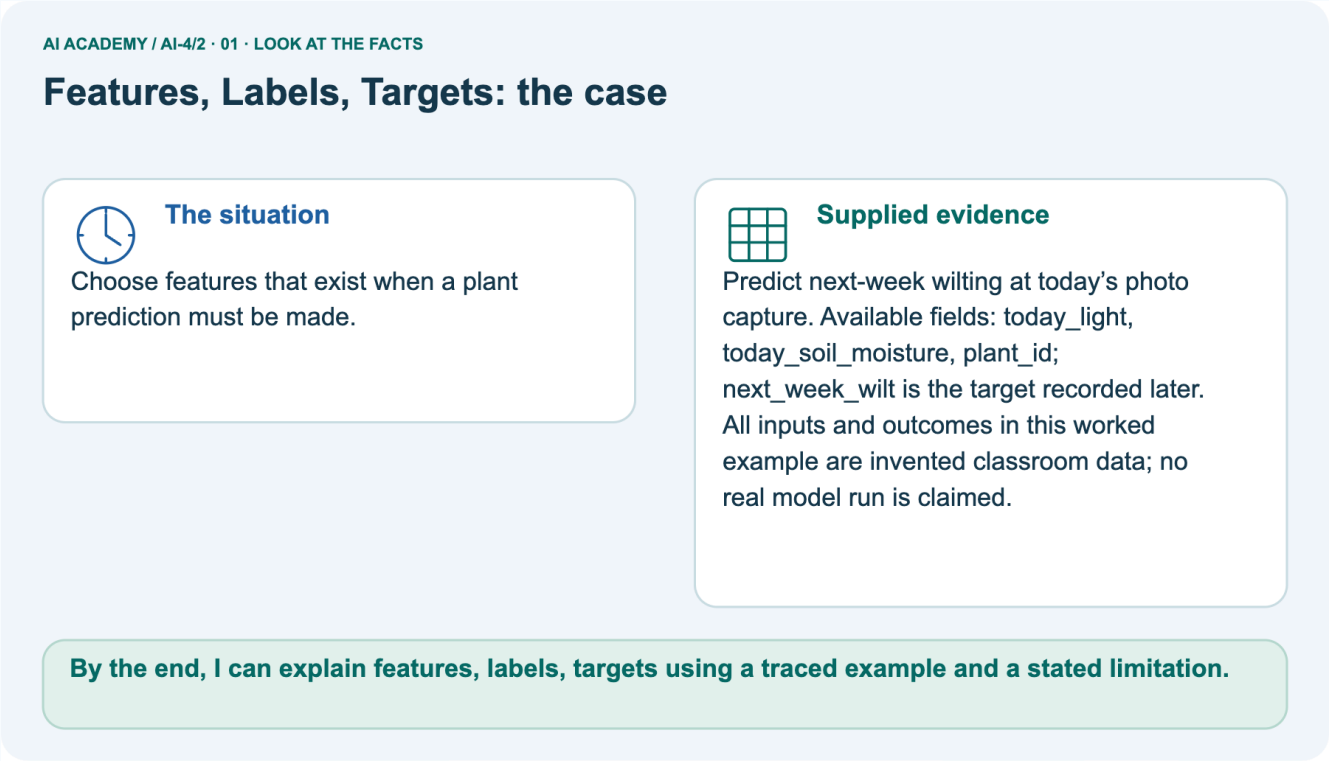
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

## Visual concept journey

> Explain each Features, Labels, Targets graphic using its labels, deciding evidence and relevant limitation.

### 01. Features, Labels, Targets: the case



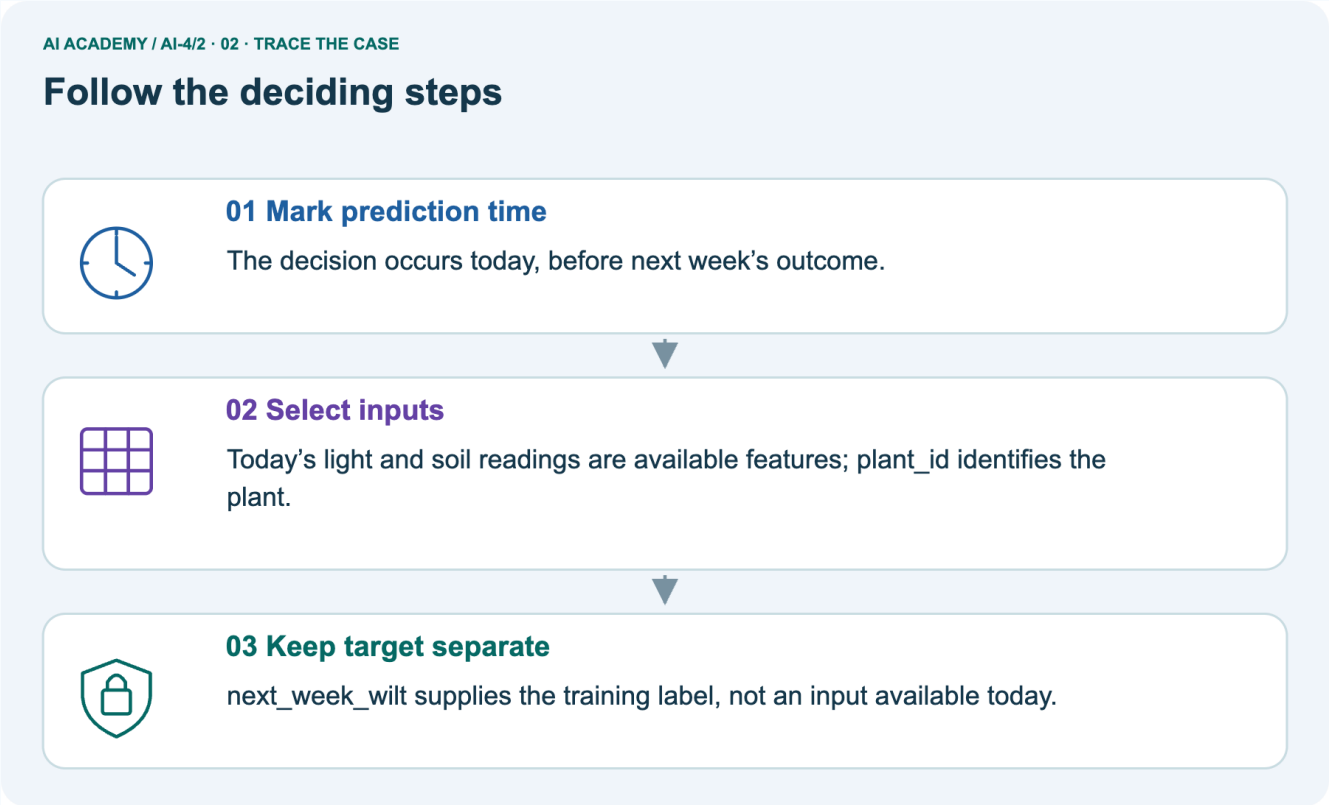

Follow the labelled picture with the matching explanation. Open the vector diagram

Choose **feature**s that exist when a plant **prediction** must be made. Supplied facts: Predict next-week wilting at today's photo capture.

Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

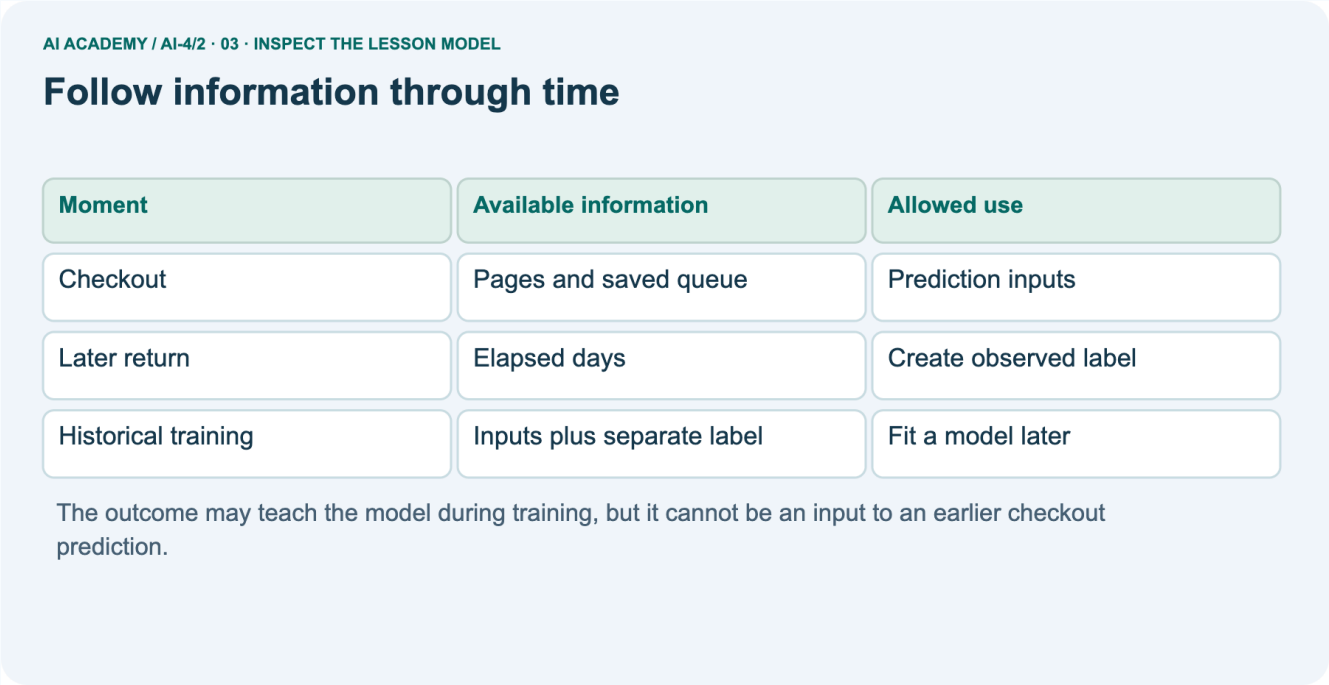



Follow the labelled picture with the matching explanation. Open the vector diagram

Mark **prediction time**: The decision occurs today, before next week's outcome. → Select **input**s: Today's light and soil readings are available **feature**s; plant_id identifies the plant. → Keep **target** separate: next_week_wilt supplies the **training label**, not an **input** available today.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. Follow information through time

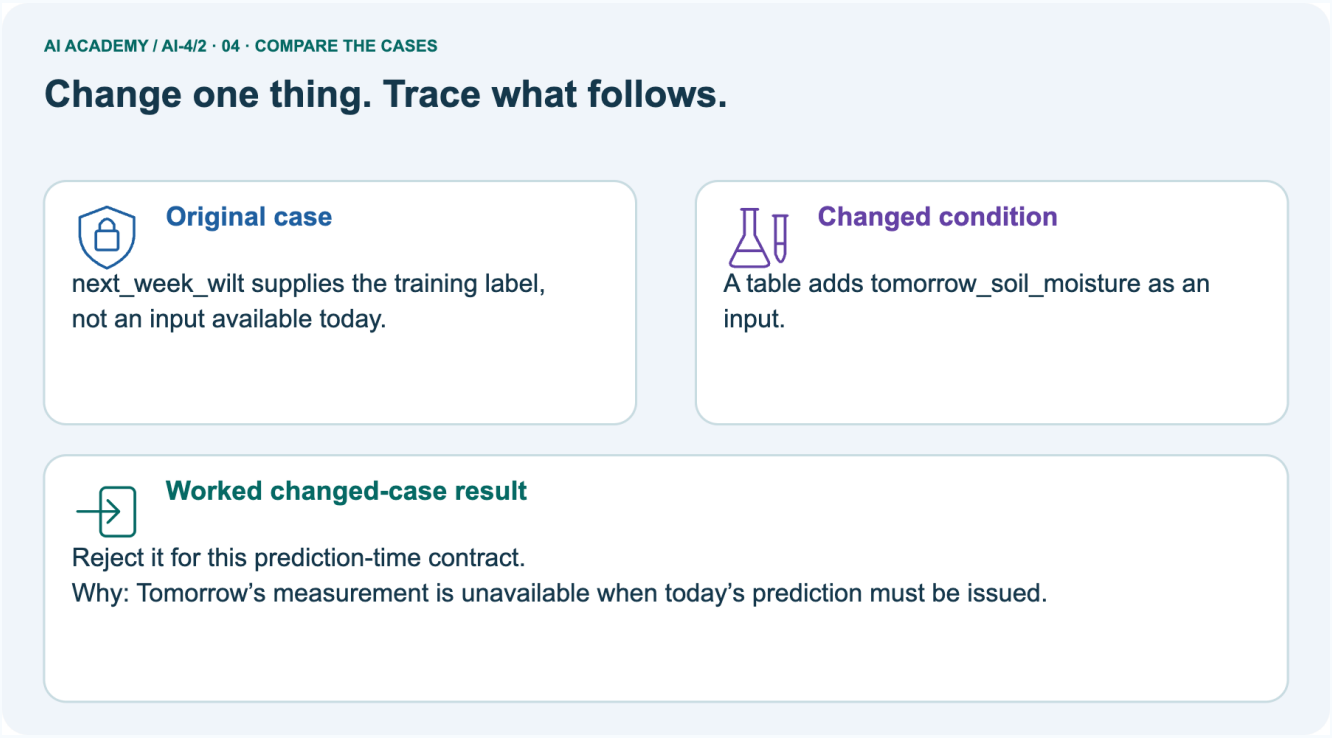



Follow the labelled picture with the matching explanation. Open the vector diagram

Follow information through time. Checkout | Pages and saved queue | **Prediction input**s. Later return | Elapsed days | Create observed **label**. Historical **training** | **Input**s plus separate **label** | Fit a model later

> Read the **label**s and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



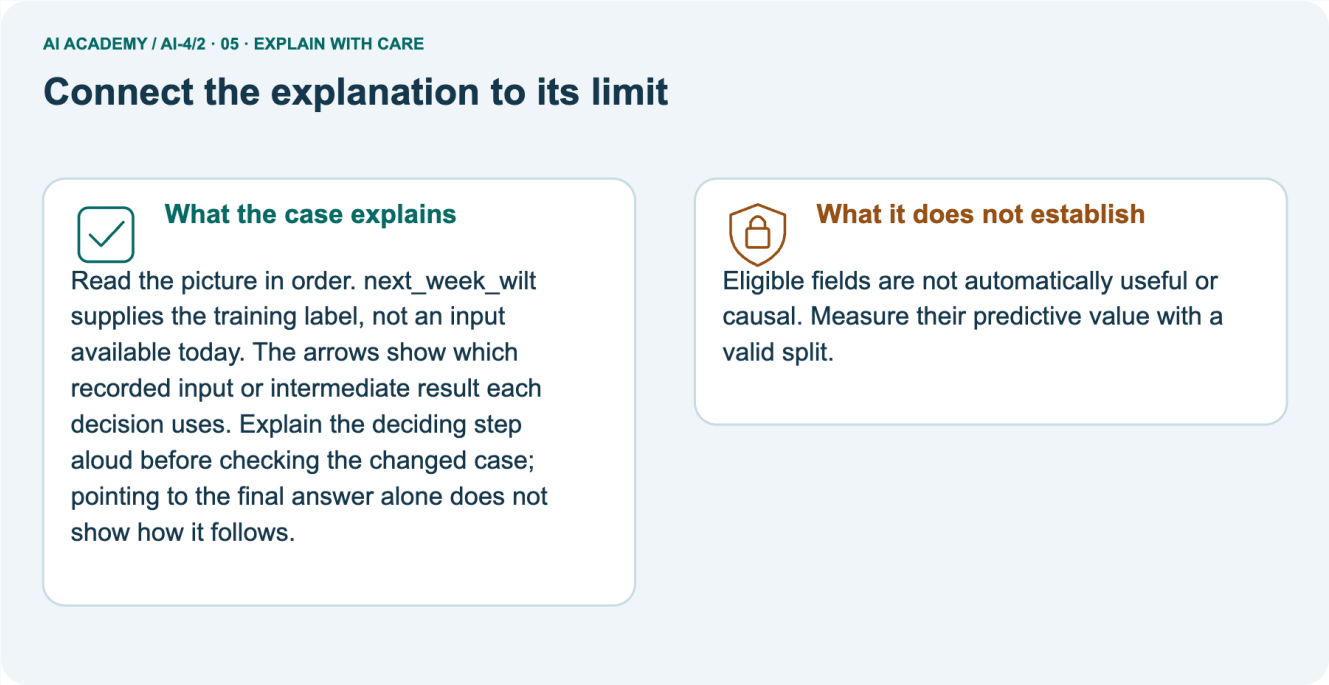

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: next_week_wilt supplies the **training label**, not an **input** available today. Changed condition: A table adds tomorrow_soil_moisture as an **input**. Expected result: Reject it for this **prediction**-time contract.

Reason: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

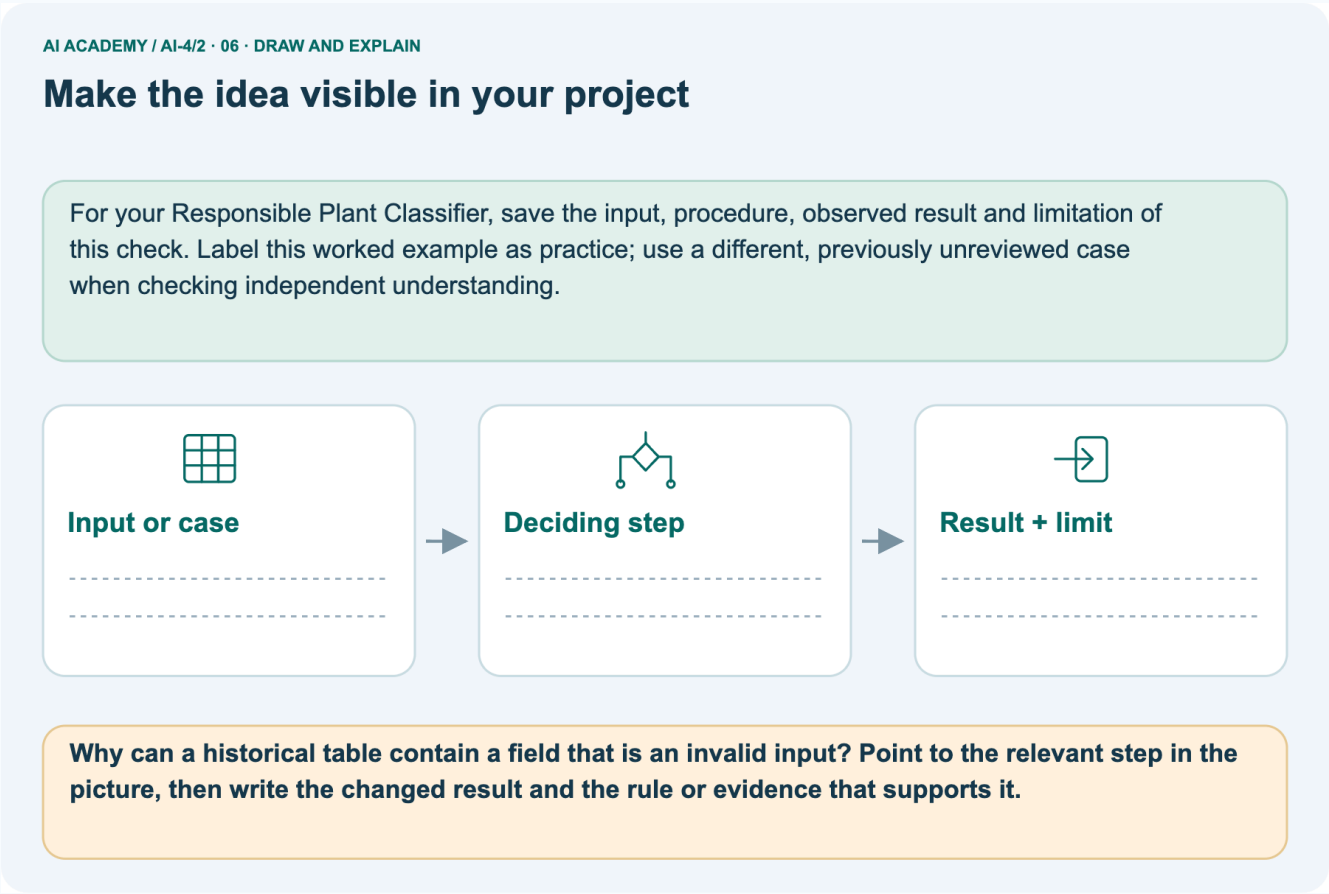
Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Eligible fields are not automatically useful or causal.

Measure their predictive value with a valid split.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

**Label** this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.